

WASP-12b

R-1144

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161208 · created 2026-07-31T00:26:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457729.75726 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.045
Transit depth (Rp/Rs)^2	1.8e-05 +/- 0.0038 [%]
Orbital Inclination (inc)	80.65 +/- 2.81
Airmass coefficient 1 (a1)	0.0371 +/- 0.0099
Airmass coefficient 2 (a2)	3.0 +/- 1.76
Scatter in the residuals of the lightcurve fit is	22.4 %
Best Comparison Star	#2 - [34, 440]
Optimal Aperture	9.0
Optimal Annulus	41.89
Transit Duration (day)	0.0029 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.045 vs 0.1178 (deviation 2.62σ)
Duration fit vs published	0.0029 d vs 0.125 d (deviation 42.1σ)
Mid-time O–C	-1571.2 min
Depth SNR	0.0
Residual scatter	22.4 %

Light curve

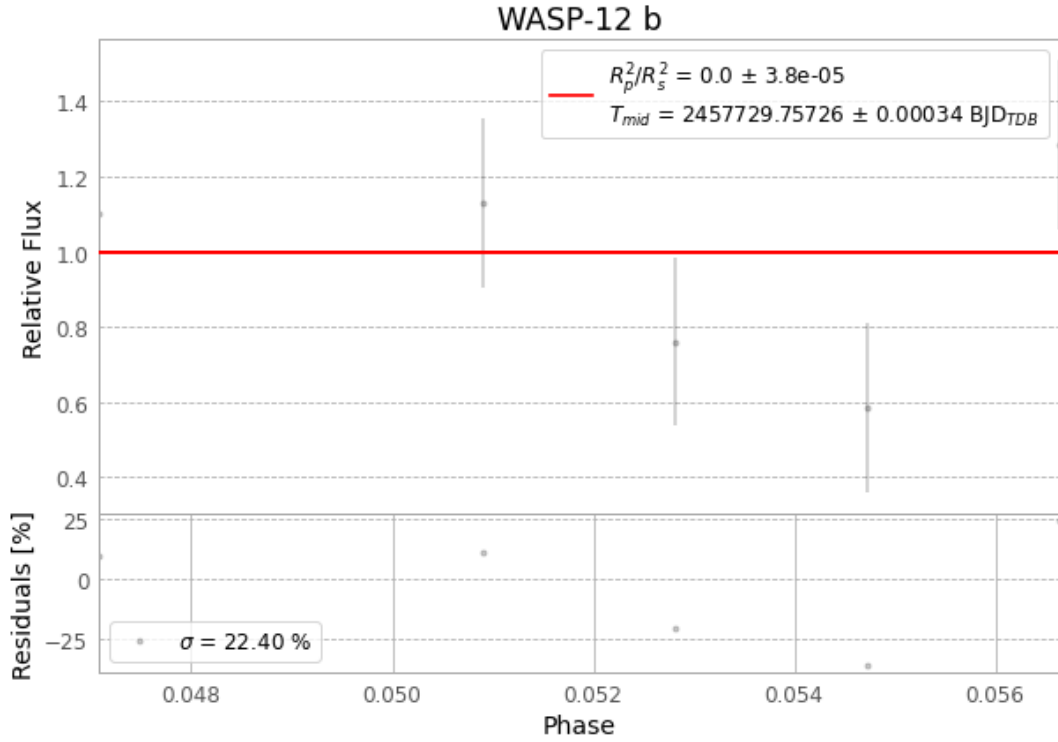


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-07.png (SHA-256 f3b4f2fba6dd3d64...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 373], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c6712dea5f02cfa3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

51 FITS frames (33 MB), WASP-12161207071212.FITS → WASP-12161207094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161208.log (206018fb314d...), FinalLightCurve_WASP-12 b_2016-12-07.png (f3b4f2fba6dd...), FinalLightCurve_WASP-12 b_2016-12-07.csv (69283e5546b7...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:26Z.